

# JEE MAIN CRASH COURSE

**IT'S TIME TO LEARN APPLICATION**

- ✓ **LATEST NTA PATTERN**
- ✓ **CONCEPT VIDEOS**
- ✓ **LIVE DOUBT SESSIONS**
- ✓ **DPP, REVISION SHEETS**
- ✓ **100% MONEYBACK TO MERITORIOUS STUDENTS**

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Designed by  
Math Maestro

**Anup Sir**



# JEE ADV. CRASH COURSE

## TARGETING JEE ADVANCED 2020

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- ✓ ADVANCED LEVEL VIDEOS
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JEE Mains 2020 Jan Chapter wise Question Bank

Ellipse

Q1

If distance between the foci of an ellipse is 6 and distance between its directrices is 12, then length of its latus rectum is

यदि एक दीर्घवृत्त के नाभियों के मध्य की दूरी 6 है तथा इसकी नियताओं के मध्य की दूरी 12 है तब इसके नाभिलम्ब की लम्बाई है—

- (1) 4 (2)  $3\sqrt{2}$  (3) 9 (4)  $2\sqrt{2}$

7<sup>th</sup> Jan Morning

Sol

(2)

$$2ae = 6$$

and और

$$\frac{2a}{e} = 12$$

$$\Rightarrow ae = 3$$

and और

$$\frac{a}{e} = 6$$

$$\Rightarrow a^2 = 18$$

$$\Rightarrow b^2 = a^2 - a^2e^2 = 18 - 9 = 9$$

$$\Rightarrow \text{L.R.} = \frac{2b^2}{a} = \frac{2 \times 9}{3\sqrt{2}} = 3\sqrt{2}$$

02

$3x + 4y = 12\sqrt{2}$  is the tangent to the ellipse  $\frac{x^2}{a^2} + \frac{y^2}{9} = 1$  then the distance between foci of ellipse is-

यदि  $3x + 4y = 12\sqrt{2}$ , दीर्घवृत्त  $\frac{x^2}{a^2} + \frac{y^2}{9} = 1$  की स्पर्श रेखा है तो दीर्घवृत्त की नाभियों के मध्य दूरी होगी—

- (1)  $2\sqrt{5}$  (2)  $2\sqrt{3}$  (3)  $2\sqrt{7}$  (4) 4

7<sup>th</sup> Jan Evening

Sol

## Ellipse

## JEE Mains 2020 Jan Chapter wise Question Bank

(3)

$$3x + 4y = 12\sqrt{2}$$

$$\Rightarrow 4y = -3x + 12\sqrt{2}$$

$$\Rightarrow y = -\frac{3}{4}x + 3\sqrt{2}$$

condition of tangency स्पर्शता का प्रतिबन्ध  $c^2 = a^2m^2 + b^2$

$$18 = a^2 \cdot \frac{9}{16} + 9$$

$$a^2 \cdot \frac{9}{16} = 9$$

$$a^2 = 16$$

$$1a = 4$$

$$e = \sqrt{1 - \frac{b^2}{a^2}} = \sqrt{1 - \frac{9}{16}} = \frac{\sqrt{7}}{4}$$

$$\therefore ae = \frac{\sqrt{7}}{4} \cdot 4 = \sqrt{7}$$

$\therefore$  focus are नाभियां  $(\pm\sqrt{7}, 0)$

$\therefore$  distance between foci नाभियों के मध्य दूरी  $= 2\sqrt{7}$

03

Ellipse  $2x^2 + y^2 = 1$  and  $y = mx$  meet a point P in first quadrant. Normal to the ellipse at P meets x-axis

at  $\left(-\frac{1}{3\sqrt{2}}, 0\right)$  and y-axis at  $(0, \beta)$ , then  $|\beta|$  is

(1)  $\frac{2}{\sqrt{3}}$

(2)  $\frac{2\sqrt{2}}{3}$

(3)  $\frac{\sqrt{2}}{3}$

(4)  $\frac{2}{3}$

8<sup>th</sup> Jan Morning

Sol

**Ellipse**

(3)

Let P be  $(x_1, y_1)$ Equation of normal at P is  $\frac{x}{2x_1} - \frac{y}{y_1} = -\frac{1}{2}$ It passes through  $\left(-\frac{1}{3\sqrt{2}}, 0\right) \Rightarrow \frac{-1}{6\sqrt{2}x_1} = -\frac{1}{2} \Rightarrow x_1 = \frac{1}{3\sqrt{2}}$ So  $y_1 = \frac{2\sqrt{2}}{3}$  (as P lies in 1<sup>st</sup> quadrant)So  $\beta = \frac{y_1}{2} = \frac{\sqrt{2}}{3}$ मनाकि बिन्दु P,  $(x_1, y_1)$  हैP पर अभिलम्ब  $\frac{x}{2x_1} - \frac{y}{y_1} = -\frac{1}{2}$ **04**

If  $e_1$  and  $e_2$  are eccentricities of  $\frac{x^2}{18} + \frac{y^2}{4} = 1$  and  $\frac{x^2}{9} - \frac{y^2}{4} = 1$ , respectively and if the point  $(e_1, e_2)$  lies on ellipse  $15x^2 + 3y^2 = k$ . Then find value of k

यदि  $e_1$  तथा  $e_2$  क्रमशः  $\frac{x^2}{18} + \frac{y^2}{4} = 1$  तथा  $\frac{x^2}{9} - \frac{y^2}{4} = 1$  की उत्केन्द्रताएँ हैं तथा बिन्दु  $(e_1, e_2)$  दीर्घवृत्त  $15x^2 + 3y^2 = k$  पर स्थित है तो k का मान है—

(1) 14

(2) 15

(3) 16

(4) 17

**9<sup>th</sup> Jan Morning****Sol**

(3)

$$e_1 = \sqrt{1 - \frac{4}{18}} = \sqrt{\frac{7}{9}} = \frac{\sqrt{7}}{3}$$

$$e_2 = \sqrt{1 + \frac{4}{9}} = \sqrt{\frac{13}{9}} = \frac{\sqrt{13}}{3}$$

$$15e_1^2 + 3e_2^2 = k \Rightarrow k = 15\left(\frac{7}{9}\right) + 3\left(\frac{13}{9}\right) \therefore k = 16$$

**05**

## Ellipse

## JEE Mains 2020 Jan Chapter wise Question Bank

Let  $x + 6y = 8$  is tangent to standard ellipse where minor axis is  $\frac{4}{\sqrt{3}}$ , then eccentricity of ellipse is

माना  $x + 6y = 8$  मानक दीर्घवृत्त की स्पर्श रेखा का समीकरण है जिसकी लघु अक्ष की लंबाई  $\frac{4}{\sqrt{3}}$  है, तो दीर्घवृत्त की उत्केन्द्रता है

(1)  $\sqrt{\frac{5}{6}}$

(2)  $\sqrt{\frac{11}{12}}$

(3)  $\frac{1}{3}\sqrt{\frac{11}{3}}$

(4)  $\frac{1}{4}\sqrt{\frac{11}{12}}$

9<sup>th</sup> Jan Evening

Sol

(2)

$$2b = \frac{4}{\sqrt{3}} \Rightarrow b = \frac{2}{\sqrt{3}}$$

$$\text{Equation of tangent} \equiv y = mx \pm \sqrt{a^2m^2 + b^2}$$

$$\text{स्पर्श रेखा का समीकरण} \equiv y = mx \pm \sqrt{a^2m^2 + b^2}$$

$$\text{comparing with } y = \frac{-x}{6} + \frac{4}{3}$$

$$\equiv y = \frac{-x}{6} + \frac{4}{3} \text{ से तुलना करने पर}$$

$$m = \frac{-1}{6} \text{ and तथा } a^2m^2 + b^2 = \frac{16}{9}$$

$$\Rightarrow \frac{a^2}{36} + \frac{4}{9} = \frac{16}{9}$$

$$\Rightarrow \frac{a^2}{36} = \frac{16}{9} - \frac{4}{9} = \frac{4}{9}$$

$$\Rightarrow a^2 = 16$$

$$e = \sqrt{1 - \frac{b^2}{a^2}}$$

$$e = \sqrt{1 - \frac{4}{3 \times 16}} = \sqrt{\frac{11}{12}}$$

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